Customer Shopping Behaviour Analysis

1. Project Overview

a. This project analyses customer shopping behaviour using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behaviour to guide strategic business decisions.

2. Dataset Summary

a. Rows: 3,900
b. Columns: 18
c. Key Features:
d. Customer demographics (Age, Gender, Location, Subscription Status)
e. Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Colour)
f. Shopping behaviour (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
g. Missing Data: 37 values in Review Rating column

3. Exploratory Data Analysis Using Python

a. We began with data preparation and cleaning in Python:

Data Loading: Imported the dataset using pandas

Initial Exploration: Used df.info( ) to check structure and .describe( ) for summary statistics.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Customer ID             3900 non-null   int64
 1   Age                     3900 non-null   int64
 2   Gender                  3900 non-null   object
 3   Item Purchased          3900 non-null   object
 4   Category                3900 non-null   object
 5   Purchase Amount (USD)   3900 non-null   int64
 6   Location                3900 non-null   object
 7   Size                    3900 non-null   object
 8   Color                   3900 non-null   object
 9   Season                  3900 non-null   object
 10  Review Rating           3863 non-null   float64
 11  Subscription Status     3900 non-null   object
 12  Shipping Type           3900 non-null   object
 13  Discount Applied        3900 non-null   object
 14  Promo Code Used         3900 non-null   object
 15  Previous Purchases      3900 non-null   int64
 16  Payment Method          3900 non-null   object
 17  Frequency of Purchases  3900 non-null   object
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB
```

	Customer ID	Age	Purchase Amount (USD)	Review Rating	Previous Purchases
count	3900.000000	3900.000000	3900.000000	3863.000000	3900.000000
mean	1950.500000	44.068462	59.764359	3.750065	25.351538
std	1125.977353	15.207589	23.685392	0.716983	14.447125
min	1.000000	18.000000	20.000000	2.500000	1.000000
25%	975.750000	31.000000	39.000000	3.100000	13.000000
50%	1950.500000	44.000000	60.000000	3.800000	25.000000
75%	2925.250000	57.000000	81.000000	4.400000	38.000000
max	3900.000000	70.000000	100.000000	5.000000	50.000000

● **Missing Data Handling**: Checked for null values and imputed missing values in the review_rating column using the median rating of each product category.

● **Column Standardization**: Renamed columns to snake case for better readability and documentation.

● **Feature Engineering**:

- Created **age_group** column by binning customer ages.
- Created **purchase_frequency_days** column from purchase data.

● Data Consistency Check: Verified if discount_applied and promo_code_used were redundant; dropped promo_code_used.

● **Database Integration**: Connected Python script to PostgreSQL and loaded the cleaned Data Frame into the database for SQL analysis.

4. Data Analysis Using SQL (Business Transactions)

We performed structured analysis in Microsoft SQL Server to answer key business questions:

1. **Revenue by Gender**: Compare total revenue generated by male vs female customers

	gender	revenue
1	Male	157890
2	Female	75191

2. **High Spending Discount Users**: Identify customers who used discounts but still spend above the average purchase amount.

	customer_id	purchase_amount
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	88
12	29	94

3. **Top 5 Products by Rating**: Find top 5 products with highest average review rating.

	item_purchased	average_product_rating
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.8
5	Skirt	3.78

4. **Shipping Type Comparison**: Compare average purchase amount between Standard and Express Shipping.

	shipping_type	average_purchase_amount
1	Standard	58.46
2	Express	60.48

5. **Subscribers VS Non-Subscribers**: Compare average spend and total revenue across subscription status.

	subscription_status	total_customers	average_spent	total_revenue
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

6. **Discount Depended Products**: Identify 5 products with the highest percentage of discount purchases.

	item_purchased	discount_rate
1	Hat	50.00
2	Sneakers	49.66
3	Coat	49.07
4	Sweater	48.17
5	Pants	47.37

7. **Customer Segmentation**: Classify customers into New, Returning and Loyal based on purchase history.

	customer_segment	number_of_customer
1	Loyal	3116
2	Returning	701
3	New	83

8. **Top 3 Products Per Category**: List most purchased products within each category.

	item_rank	category	item_purchased	total_orders
1	1	Accessories	Jewelry	171
2	2	Accessories	Belt	161
3	3	Accessories	Sunglasses	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

9. **Repeat Buyers and Subscription**: Check weather buyers with more than 5 purchases are more likely to subscribe.

	subscription_status	repeat_buyers
1	Yes	958
2	No	2518

10. **Revenue by Age Group**: Calculate total revenue contribution of each age group.

	age_group	total_revenue
1	Young Adult	62143
2	Middle Aged	59197
3	Adult	55978
4	Senior	55763

5.Dashboard in Power BI

Finally, we build a dashboard in **Power BI** to present the insights visually.